# HAFEEFA P C

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## Professional Summary

AI/ML Engineer with experience developing LLM-powered agents, evaluating model reliability, and building AI-driven applications. Interested in Generative AI, retrieval-augmented generation (RAG), machine learning, and AI safety, with additional experience in full-stack development and data processing pipelines.

## Experience

**Tata Consultancy Services (TCS)** Nov 2025 – Present
*AI/ML Automation Engineer* *Kochi, India*
- Develop and evaluate LLM-powered agents, focusing on reliability, safety, alignment, and robust behavior.
- Conduct empirical analysis of agent behavior, hallucinations, adversarial robustness, and failure mitigation to support measurable AI evaluation and improvement.
- Contribute to a Google Cloud Dataflow ingestion proof of concept, covering data validation, normalization, error handling, logging, and pipeline metrics.

**Kireap** Aug 2024 – Jan 2025
*Full Stack Developer Intern* *Bangalore, India*
- Contributed to the Kireap Flutter application to improve the client onboarding experience.
- Integrated backend APIs and performed Postman-based testing to identify and resolve integration issues.
- Researched token systems and blockchain traceability, proposing improvements to data transparency and security.

**Xplor** Dec 2023 – Feb 2024
*Frontend Developer Intern* *Kochi, India*
- Developed interactive website components using React.js and integrated REST APIs to support application functionality.
- Collaborated with backend developers to improve data flow and page responsiveness, contributing to a reported 15% improvement.
- Validated test cases and supported frontend debugging and testing.

## Selected Projects

**LifeScience-AI Agent** | *Python, Generative AI, RAG, LLMs*
- Developed an AI-powered research assistant for answering life-sciences questions using curated biomedical literature and regulatory documents.
- Implemented a RAG pipeline incorporating document chunking, embeddings, vector retrieval, context selection, and answer synthesis to ground LLM responses in source material.
- Optimized prompting and context selection, with caching strategies to improve response efficiency and reduce hallucinations during evaluation.

**Credit Card Fraud Detection** | *Python, Scikit-learn, XGBoost*
- Built a machine learning pipeline to identify fraudulent transactions in a dataset containing over 284,807 records with highly imbalanced classes.
- Compared Logistic Regression, Random Forest, and XGBoost using SMOTE, random undersampling, and class weighting to address class imbalance.
- Applied threshold optimization to achieve a reported PR-AUC of 0.88 and F1-score of 86%; also explored PCA and KMeans/DBSCAN for dimensionality reduction and transaction-pattern analysis.

**Aurum-Chain** | *Next.js, Solidity, Scaffold-ETH 2, Tailwind CSS* GitHub
- Developed a blockchain-powered gold traceability platform to track gold from mine to market, improving transparency and targeting fraud prevention.
- Built a responsive interface using Next.js, Tailwind CSS, and Shadcn UI, with interactive dashboards for customers, suppliers, and administrators.
- Implemented Solidity smart contracts to register, track, and transfer gold ownership, supporting secure and transparent ownership records.
- Explored privacy-preserving identity verification using Anon Aadhaar and decentralized storage with Arweave.

## Education

**Cochin University of Science and Technology (CUSAT)** Nov 2021 – Apr 2025
*B.Tech. in Information Technology | GPA: 9.6/10* *Kochi, India*

**PKMM Higher Secondary School** Jun 2019 – Mar 2021
*Higher Secondary Education, Science | 98.5%* *Kerala, India*

## Achievements

- **Winner – ETHIndia 2024**, an Ethereum-focused hackathon.
- **Winner – BuildOnChain 2024**, a Web3 hackathon hosted by LayerEdu at NIT Calicut.
- Recognized among the **Top 100 coders** by the MuLearn Community during Huddle Global 2023.

## Technical Skills

**Languages:** Python, Java, JavaScript, TypeScript, SQL, C/C++, Dart
**AI/ML:** Generative AI, Large Language Models, RAG, Prompt Engineering, Model Evaluation, Scikit-learn, XGBoost, PCA, SMOTE
**Frameworks:** React.js, Next.js, Node.js, Flutter, Tailwind CSS
**Data and Cloud:** MySQL, PostgreSQL, Firebase, Google Cloud Dataflow, REST APIs
**Tools:** Git, GitHub, Postman, Figma